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Llama 2 Prompt Engineering — Extracting Information From Articles Examples



Eliran Boraks · Follow

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Intro

When I started working on Llama 2, I googled for tips on how to prompt it. I quickly discovered the information was sparse and inconsistent, so I experimented and played around. I wanted to give back and share what I had learned.

What I learned is that prompt engineering is a combination of art and science. As Thomas Edison said about prompt engineering, *"I have not failed. I've just found 10,000 ways that won't work."* This pretty much encapsulates it.

This post is focused on information-extracting use cases, but the lessons can easily be applied to other use cases. I used a subset of the text from Andreessen Horowitz's blog about [Play to Win: Mobile Game Soft Launch Best Practices](#) by [Doug McCracken](#) and [Joshua Lu](#).

This post summarizes the experiments documented in Colab [prompt_engineering_experiments_11_23.ipynb](#). To run this notebook, you will need HuggingFace Pro and an auth token.

What would you learn?

- The importance of the system prompt — Giving Llama context and persona to solve a problem.
- Chain-of-Thought prompting — Customize a prompt using answers from the previous prompt
 - Output an answer in a valid JSON
- One-to-Many Shot Learning — Teach Llama how to solve a problem with examples.

The Importance of the System Prompt

Before introducing the system prompt, let's use the simple prompt to summarize the article into bullet points.

```
p1 = """Write a concise summary of the main ideas in article below in bullet-points"""  
  
display_answer(run_prompt(p1))
```

```
Do players understand how to play the game?  
A game's first tutorial is a great place to start learning, but it's important to  
How much can you spend for user acquisition?  
You can use the soft launch period to see how your user acquisition strategy works  
How much do players monetize?  
You may want to know how much players will spend on in-game items and currency to  
How does the game stand out from the competition?  
It's important to know how your game compares to other games in the same genre,  
How to Soft Launch  
Now that we've established what a soft launch is, why it's important, and what you  
Step 1: Choose Your Market  
The first step is choosing which market you want to soft launch in. You want to
```

Step 2: Set Your Testing Goals

Next, you want to set specific goals for what you want to test during the soft launch.

Step 3: Choose Your Test Size

The next step is to decide on the size of your test. This will depend on the method you choose.

Step 4: Choose Your Test Period

Next, you want to decide on the length of your soft launch test. This will depend on the size of your test.

Step 5: Plan Your User Acquisition Strategy

During the soft launch, you want to focus on acquiring players who are similar to your target audience.

Step 6: Monitor and Analyze Your Metrics

Once your soft launch is under way, you want to closely monitor your metrics and analyze the data.

Step 7: Iterate and Refine

After you have gathered all the data you need from your soft launch, it's time to iterate and refine your strategy.

Step 8: Set a Launch Date

Finally, once you are happy with the changes you've made, it's time to set a launch date.

Not bad for our first attempt. However, the response is too long. It's a rewrite of the article instead of a summary, and Llama ignored the idea of bullet points.

I tried to improve the results by asking Llama for TL;DR, not repeating points, and limiting the number of points to five. As you can see below, it worked somewhat, but I can't call it success. Next, add context via the system prompt.

```
p3 = """Write a concise TL;DR summary in numeric bullet-points for the following article. Don't repeat ideas in bullet points. Limit the number of bullet points to 5. article_text = """
display_answer(run_prompt(p3))
```

- Do players find the gameplay compelling?

In addition to stability, you should test whether the gameplay is compelling enough.

- Can you generate revenue?

In order for a game to succeed, it needs to be able to generate revenue. The soft launch is a good time to test this.

- Is the game's meta loop strong enough to attract and retain players?

The meta gameplay loop is the broader loop of how players interact with your game.

- Can you acquire users at a reasonable cost?

Acquiring users at a reasonable cost is important for the long-term success of the game.

What are the benefits of a soft launch?

A soft launch is an opportunity to test your game with a limited number of players.

Some of the benefits of a soft launch include:

- Testing gameplay and graphics: A soft launch allows you to test the gameplay and graphics with a limited number of players.
- Gathering player feedback: A soft launch is an opportunity to gather feedback from players.
- Monitoring player behavior: A soft launch allows you to monitor player behavior and make adjustments.

- Improving monetization: A soft launch is a chance to test your monetization strategy.
- Building a community: A soft launch is an opportunity to build a community of players.
- Reducing risk: A soft launch can help you reduce the risk of launching a game.

TL;DR:

1. Soft launches are a way to test a game with a limited number of players before a full launch.
2. They allow developers to gather feedback, monitor player behavior, and identify issues.
3. Soft launches can help developers improve monetization, build a community, and reduce risk.
4. There are two main stages of soft launch: alpha and beta.
5. Alpha is used to test the core gameplay loop, while beta is used to test the monetization strategy.
6. Soft launches can be beneficial, but they are not necessary for a game to succeed.
7. It's important to choose a market that closely aligns with the target demographic.
8. Soft launches can be done on PC and console games as well, not just mobile games.
9. Resetting player progress after a soft launch can be done, but developers should be careful.

Llama 2's System Prompt

Llama 2 was trained with a system message that set the context and persona to assume when solving a task. For more information, see the [system message described on the Hugging Face blog](#).

We will start by adding the standard system message from the Hugging Face blog to the p3 prompts from above.

```
p4 = """<s>[INST] <<SYS>>
You are a helpful, respectful and honest assistant. Always answer as helpfully as you can.
Your answers should not include any harmful, unethical, racist, sexist, toxic, dangerous, or illegal content.
Please ensure that your responses are socially unbiased and positive in nature.
If a question does not make any sense, or is not factually coherent, explain why.
If you don't know the answer to a question, please don't share false information.
<</SYS>>
Write a concise TL;DR summary in numeric bullet-points for the following article.
don't repeat ideas in bullet points. Limit the number of bullet-point to 5. article:
How to Soft Launch a Game
Here are some tips for soft launching a game:
1. Start small: Don't launch to too many players at first, as this will make it difficult to manage.
2. Choose a small, representative market: Choose a market that closely aligns with your target audience.
3. Set up analytics: Make sure you have analytics set up to track player behavior and engagement.
4. Test different aspects of the game: Use the soft launch to test different aspects of the game, such as monetization, player progression, and social features.
5. Gather feedback: Listen to player feedback and use it to improve the game before the full launch.
</INST>"""

p4_answer = run_prompt(p4)
display_answer(p4_answer, "P4 - With System Message")
```

Do players find the game fun and engaging?
 It is a best practice to test the retention rate of the game during the soft launch.
 Do players understand the monetization strategy?
 It is a good idea to soft launch your game to test the monetization strategy. The goal is to see if players can effectively acquire users.
 The cost of acquiring users can vary greatly by region and country and soft launch can help you understand this better.
 How to Soft Launch a Game
 Here are some tips for soft launching a game:
 1. Start small: Don't launch to too many players at first, as this will make it difficult to manage.
 2. Choose a small, representative market: Choose a market that closely aligns with your target audience.
 3. Set up analytics: Make sure you have analytics set up to track player behavior and engagement.
 4. Test different aspects of the game: Use the soft launch to test different aspects of the game, such as monetization, player progression, and social features.
 5. Gather feedback: Listen to player feedback and use it to improve the game before the full launch.

5. Be transparent with players: Be upfront with players that the game is in a soft launch phase.
 6. Keep the soft launch exclusive: Make sure that the soft launch is invite-only.
 7. Respect player feedback: Take player feedback seriously and use it to improve the game.
 8. Don't be afraid to pivot: If the soft launch reveals that the game is not working, be prepared to pivot.
 9. Don't forget to localize: If you plan to release your game in multiple countries, make sure to localize the game.
 10. Don't rush the launch: Take the time to properly test and refine the game before launching.
- In conclusion, soft launching a game can be a valuable tool for game studios. It

Out of the box, the system prompt improved the answer, but it's still too wordy and doesn't provide a concise overview of the article. Still, the use of bullet points is not what I was expecting.

Customize the System Prompt

The nice thing about Llama (compared to ChatGPT) is that we have more control over the system prompt. The following experiment will instruct Llama to assume the persona of a researcher tasked with writing a concise brief. Changes applied to the original system prompt:

- Use the researcher persona and specify the tasks to summarize articles.
- Remove safety instructions; they are unnecessary since we ask Llama to be truthful to the article.

```
p5 = """<s>[INST] <<SYS>>
You are a researcher task in summarizing and writing concise brief of articles.
Please ensure that your responses are socially unbiased and positive in nature.
If a question does not make any sense, or is not factually coherent, explain why
If you don't know the answer, please don't share false information.
<</SYS>>
Write a concise TL;DR summary in numeric bullet-points for the following article
don't repeat ideas in bullet points. Limit the number of bullet-point to 5. arti
[/INST]"""
format(BODY=text)

display_answer(run_prompt(p5))
```

Here are five key points from the article "Play to Win: Mobile Game Soft Launch

1. Soft launching a mobile game allows developers to test different aspects of the game.
2. A soft launch typically involves releasing the game to a limited number of players.
3. There are two stages of soft launch: alpha and beta. Alpha testing is used to test the game with a small group of players.
4. Soft launching is important because it allows developers to test the game with real players before a full launch.
5. Some common myths about soft launching include the idea that the country or market is not ready for the game.

Wow!!! Now things are looking better. I got five bullet points that summarize the article. Also, this response took Llama a third of the time compared to earlier experiments — time is money!!!

Chain-of-Thought prompting

Chain-of-thought is when a prompt is being constructed using a previous prompt answer. For our use case to extract information from text, we will first ask Llama what the article is about and then use the response to ask a second question: what problem does [what the article is about] solve?

The following code has two prompts. P7 asks Llama what the article is about, and the answer is then used in a second prompt: what problem does [answer to 1st prompt] solve? To make it easy to work with the response to the first prompt, I asked Llama to return the answer in JSON.

```
p7 = """<s>[INST] <<SYS>>
You are a researcher task with answering questions about an article.
Please ensure that your responses are socially unbiased and positive in nature.
If a question does not make any sense, or is not factually coherent, explain why
If you don't know the answer, please don't share false information.
<</SYS>>
Tell me what is the article about in one to three words?
Output the answer in JSON in the following format {"article_is_about": [answer_
Article: {BODY}
[/INST]""".format(BODY=text)

a7 = run_prompt(p7)
json_a7 = json.loads(a7.answer)
print(f"JSON Answer:\n {json_a7}")
about = json_a7['article_is_about']

p8 = """<s>[INST] <<SYS>>
You are a researcher task with answering questions about an article.
Please ensure that your responses are socially unbiased and positive in nature.
If a question does not make any sense, or is not factually coherent, explain why
If you don't know the answer, please don't share false information.
<</SYS>>
According to the article, what problems does {{ABOUT}}? Limited the answer to 20
Article: {BODY}
[/INST]""".format(BODY=text, ABOUT=about)

display_answer(run_prompt(p8), 'Prompt 8')
```

This is pretty sweet; I got a concise list of problems that mobile game soft launch is solving.

```
JSON Answer:
{'article_is_about': 'mobile game soft launch best practices'}

According to the article, soft launching a mobile game can help studios:

* Test the game's stability and performance in a real-world environment
* Gather player feedback and metrics to improve the game before global release
* Understand the game's potential for success and identify areas for improvement
* Save money by identifying and fixing issues before a full release
* Avoid angering the community by resetting player progress and being upfront ab
```

Generating JSON Tips

Llama needs precise instructions when asking it to generate JSON; the Colab notebook [prompt_engineering_experiments_11_23.ipynb](#) goes into it in more detail. In essence, here is what works for me to get valid JSON consistently:

1. Explicitly state — “All output must be in valid JSON. Don’t add explanation beyond the JSON” in the system prompt.
2. Add an “explanation” variable to the JSON example. Llama enjoys explaining its answers. Give it an outlet.
3. Use the JSON as part of the instruction. See the “in_less_than_ten_words” example below.
4. Change “write the answer” to “output the answer.”

Here is an example prompt asking for JSON output.

```
<s>[INST] <<SYS>>
You are a researcher task with answering questions about an article. All output
Please ensure that your responses are socially unbiased and positive in nature.
If a question does not make any sense, or is not factually coherent, explain why
If you don't know the answer, please don't share false information.
<</SYS>>
What sport is the article about? Output must be in valid JSON like the following
Article: {BODY}
[/INST]
```

One-to-Many Shot Learning | Lead by Example

In this last example, I want Llama to identify general topics in the text. For example: what industry, sport, and company are mentioned in the article? However, the problem with this approach is that it is hard to scale. An

unlimited number of topics/ideas can be said in articles, and trying to predict them all defeats the purpose of using AI. Instead, I am going to use a technique called One-to-Many Shot Learning.

One-to-Many Shot Learning is a term that refers to a type of machine learning problem where the goal is to learn to recognize many different classes of objects from only one or a few examples of each class. For example, if you have only one image of a cat and one image of a dog, can you train a model to distinguish between cats and dogs in new images? This is a challenging problem because the model has to generalize well from minimal data ([source](#))

The following prompt gives Llama examples of the type of topic I am looking for and asks it to find a similar subject in the article.

Important points about the prompts:

1. The system prompt includes the instructions to output the answer in JSON.
2. The prompt consists of an one-to-many shot learning section that starts after `<</SYS>>` and ends with `</s>`. *See the prompt template below will make it easier to understand.*
3. The examples are given in JSON because the answers need to be JSON.
4. The JSON allows defining the response with *name*, *type*, and *explanation*.

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One-to-Many Shot Learning Prompt Template

```
<s>[INST] <<SYS>>
SYSTEM MESSAGE
<</SYS>>

EXAMPLE QUESTION [/INST]
EXAMPLE ANSWER(S)
</s>

<s>[INST]
QUESTION
[/INST]
```


One-to-Many Shot Learning Example

```
p14 = """<s>[INST] <<SYS>>
You are a researcher tasked with answering questions about an article.
Please ensure that your responses are socially unbiased and positive in nature.
If a question does not make any sense or is not factually coherent, explain why
If you don't know the answer, please don't share false information.

Output answer in JSON using the following format: {"name": name, "type": type,
<</SYS>>

What nouns are mentioned in the article that can generalize the topic? [/INST]
[
{"name": "semiconductor", "type": "industry", "explanation": "Companies engaged
{"name": "NBA", "type": "sport league", "explanation": "NBA is the national bas
{"name": "Ford F150", "type": "vehicle", "explanation": "Article talks about th
{"name": "Ford", "type": "company", "explanation": "Ford is a company that buil
{"name": "John Smith", "type": "person", "explanation": "Mentioned in the artic
] </s>

<s>[INST]
What nouns are mentioned in the article that can generalize the topic? Output an
Article: {BODY}
[/INST]""".format(BODY=text)

display_answer(run_prompt(p14))
```

One-to-Many Shot Learning Response

Wow!!! The response is pretty awesome. Llama identified “mobile game” as an industry and the “Apple App Store” as a platform. Giving the Llama example, is a powerful technique.

```
[
  {
    "name": "mobile game",
    "type": "industry",
    "explanation": "The article discusses the soft launch of mobile games"},
  {
    "name": "soft launch",
    "type": "term",
    "explanation": "A limited release of a game to a small number of players bef
  {
    "name": "geo test",
    "type": "term",
    "explanation": "Another name for a soft launch, used to test the game in a s
  {
    "name": "beta",
    "type": "term",
    "explanation": "A later stage of soft launch, used to test the game's meta g
  {
    "name": "alpha",
```

```

"type": "term",
"explanation": "An early stage of soft launch, used to test the game's core
{
  "name": "Apple App Store",
  "type": "platform",
  "explanation": "A digital distribution platform for iOS apps and games"},
{
  "name": "Google Play Store",
  "type": "platform",
  "explanation": "A digital distribution platform for Android apps and games"}
{
  "name": "Supercell",
  "type": "company",
  "explanation": "A mobile game development company that has used soft launch
{
  "name": "Flood Rush",
  "type": "game",
  "explanation": "A mobile game that was tested through a limited open beta be
{
  "name": "QA team",
  "type": "term",
  "explanation": "A team responsible for testing and quality assurance in game
{
  "name": "lower-spec Android phones",
  "type": "device",
  "explanation": "Lower-end Android devices that may have different performanc
{
  "name": "FPS",
  "type": "term",
  "explanation": "Frames per second, a measure of a game's performance"},
{
  "name": "crash rates/DAU",
  "type": "term",
  "explanation": "Crash rates per daily active user, a measure of a game's sta
{
  "name": "crashes/crasher",
  "type": "term",
  "explanation": "The number of crashes per user, a measure of a game's stabil
{
  "name": "latency",
  "type": "term",
  "explanation": "The delay between a user's action and the game's response, a
]

```

Conclusion

Prompt engineering requires creativity and perseverance to get it right. The lessons learned from the experiments are as follows:

- Customizing the system message to the tasks is crucial, and it is worth experimenting with and adjusting it to fit each task.
- One-to-many shot learning should be one of the first things to try to improve the response (after the system message).
- Chain-of-thought prompting is a powerful technique to uncover better responses.

- Prompt engineering is a combination of art and science. Don't give up; keep experimenting.

I hope these examples gave you ideas and inspiration.

Prompt Engineering

Llama 2

Hugging Face



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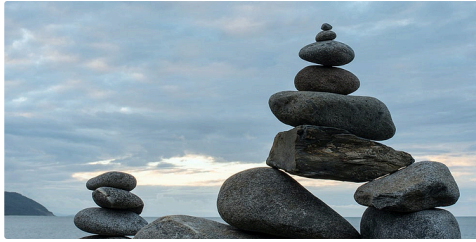
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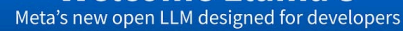
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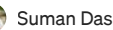
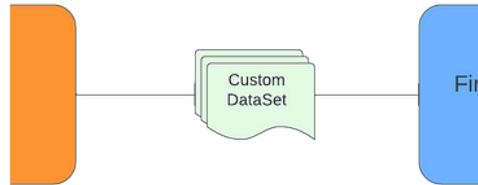
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